Neiv Gupta

neivgupta.dev | (669) 238-1070 | neiv06@ucla.edu | linkedin.com/in/neiv-gupta | github.com/neiv06

Education

University of California, Los Angeles — Los Angeles, CA
Bachelor of Science, Computer Science — *Expected June 2028*

- **Relevant Coursework:** Algorithms & Complexity, Computer Systems Architecture, Software Construction, Electrical & Electronic Circuits, Operating Systems, Logic Design of Digital Systems, Discrete Structures

Experience

Software Engineering Intern — June 2026 – Present
Pairwise Technologies (SaaS Startup) — *Los Angeles, CA*

- Built backend services with Django ORM models, DRF API endpoints, and Celery tasks for async processing.
- Developed a dedup pipeline scoring semantic cosine & lexical trigram similarity, re-ranked for diversity via MMR.
- Engineered an MCP server with FastMCP, Django, and OAuth CIMD, enabling Claude Connector workflows.

Software Engineering Intern — Apr. 2025 – Sep. 2025
ThinkScan Technologies (AI Startup) — *Pleasanton, CA*

- Developed an object detection and scene reasoning AI Agent for threat detection in defense field deployments.
- Built multi-modal retrieval system over FAISS vector store for search across CLIP-embedded imagery datasets.
- Implemented Laplacian variance and bilateral filtering to enhance noisy field imagery for inference.

Research

Undergraduate Student Researcher — May 2026 – Present
UCLA Computational Machine Learning Lab, Prof. Cho-Jui Hsieh — *Los Angeles, CA*

- Researching current audio-LLM defense system robustness against harmful queries and adversarial attacks.
- Engineering a PGD-optimized adversarial audio perturbation that bypasses Qwen2-Audio safety refusals.
- Extending gradient-direction analysis of attack method into the foundation for a new attack-aware defense.

Computer Vision Student Researcher — June 2024 - Aug 2024
Argonne National Laboratory — *Lemont, IL*

- Deployed TensorFlow/PyTorch CNN pipelines for wildfire response and drought monitoring.
- Built CLIP-based zero-shot classification model achieving 93% mAP for wildfire and drought imagery datasets.
- Fine-tuned OpenCLIP ViT-B/32 multi-modal models on imagery datasets for improved accuracy.

Projects

CudaFire | *CUDA, C++17, C, CMake, GDAL, OpenGL* — Dec. 2025 – Jan. 2026

- Built GPU-accelerated wildfire spread simulator using CUDA and the Rothermel fire behavior model.
- Processed 8.7 million terrain cells in parallel using 8-connected cellular automaton on RTX 3080.
- Optimized CUDA kernels with 16×16 thread blocks, achieving 7,643× real-time simulation performance.

BruinMarket | *Go, React, PostgreSQL, WebSockets, Docker, Railway, Vercel* — Oct. 2025 – Nov. 2025

- Launched full-stack UCLA-exclusive student marketplace with real-time peer-to-peer transactions and messaging.
- Architected backend using Go with PostgreSQL database and JWT authentication and email verification.
- Deployed production app on Railway and Vercel with custom domain configuration and CI/CD pipeline.

YUM | *React Native (Expo), Node.js, Express.js, MongoDB Atlas, REST APIs* — Mar. 2025 – Jun. 2025

- Developed UCLA mobile dining app providing live dining updates, commenting workflows, and personal profiles.
- Architected full-stack MERN mobile app with JWT-based authentication and real-time state synchronization.
- Implemented RESTful API endpoints with Express.js middleware and MongoDB aggregation pipelines.

Stairmasters | *Swift* — Aug 2024 – Dec. 2024

- Developed Swift iOS accessibility app helping UCLA students with disabilities find accessible campus routes.
- Leveraged Apple's MapKit framework with MKDirections API for accessible route calculations and navigation.
- Mapped elevator access points using Swift Core Location framework for wheelchair-accessible campus navigation.

Technical Skills

Languages: Java, Python, C, C++, Go, JavaScript, TypeScript, HTML/CSS, XML, JSON, Swift, Bash, SQL, CUDA
Frameworks: PyTorch, TensorFlow, LangChain, React, React Native, Node.js, Express.js, Gin, Django, Celery
Dev. Tools: Git, Github, Docker, Vercel, REST APIs, WebSockets, MongoDB, PostgreSQL, OpenGL, FastMCP
Libraries: NumPy, pandas, Matplotlib, Scikit-learn, OpenCV, CLIP, OpenCLIP, YOLO, XGBoost, Tailwind CSS